ENES 246 · DIGITAL LOGIC DESIGN

Lecture 28

Final Exam Preparation: Memory and PLDs

Memory sizing and expansion · timing · Hamming codes · ROM, PLA and PAL · final exam scope and study plan



Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapter 7 and course review

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Today’s roadmap

1 §7.2

Memory

Sizing, expansion, timing

2 §7.4

Error correction

Hamming encode and fix

3 §7.6

PLDs

ROM, PLA, PAL

4

The final

Scope and study plan

By the end of class you can…

- Size and expand memories and assign address ranges
- Compute clock periods per memory access
- Encode data and correct an error with a Hamming code
- Choose true or complemented outputs to share PLA terms
- Plan your study for the comprehensive final

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Agenda for today’s session



Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome; Homework 11 collected	—
0:05 – 0:30	Sizing, expansion and timing	Reviews 1, 2
0:30 – 0:50	Error correction	Review 3
0:50 – 1:00	Break	
1:00 – 1:25	ROM, PLA and PAL	Reviews 4, 5
1:25 – 1:40	Timed practice set	Practice set
1:40 – 1:50	Final exam scope and study plan	—

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Review 1 — Memory expansion

§7.3

REVIEW

Build a 1K × 16 memory from 256 × 8 chips. Chips, address lines and decoder? Which bank holds address 600?

Bank	A_9A_8	Addresses	Chips
0	00	0–255	2 (low, high byte)
1	01	256–511	2 (low, high byte)
2	10	512–767	2 (low, high byte)
3	11	768–1023	2 (low, high byte)

1. Width: $16/8 = 2$ chips side by side
2. Depth: $1024/256 = 4$ banks → **8 chips**
3. $1K = 2^{10}$: 10 lines; $256 = 2^8$: each chip takes A_7–A_0
4. A_9A_8 → a **2-to-4** decoder; each output selects one bank
5. $600 = 512 + 64 + 16 + 8 = 10\,0101\,1000_2$
6. $A_9A_8 = 10$ → **bank 2** (addresses 512–767)

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Review 2 — Timing and cells

§7.2

REVIEW

(a) An 80 MHz CPU reads a 45 ns memory. Clock periods per read? (b) SRAM or DRAM for each use?

1. (a) $T = 1/80\ \text{MHz} = 12.5$ ns
2. $45/12.5 = 3.6$ → round up: **4 clock periods** (3 wait states)
3. Access time: address to data; cycle time: between operations
4. (b) Fast, small → SRAM; dense, cheap → DRAM; kept at power-off → flash

DRAM stores charge on a capacitor that leaks: every row is refreshed within about 64 ms. SRAM is a latch: no refresh.

Use	Choice
CPU cache	SRAM: speed
Main memory	DRAM: density, cost
FPGA block RAM	SRAM
Program storage	flash (nonvolatile)

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Review 3 — Hamming code

§7.4

REVIEW

Encode 1011 0101 with a Hamming code. The word arrives with bit 10 flipped: show the correction.

1	2	3	4	5	6	7	8	9	10	11	12
0	0	1	1	0	1	1	0	0	1	0	1

0	0	1	1	0	1	1	0	0	0	0	1

received (bit 10 flipped)

1. 8 data bits need 4 check bits ($2^4 \geq 8 + 4 + 1$)
2. Data 1011 0101 into positions 3, 5, 6, 7, 9, 10, 11, 12
3. $P_1 = 1 \oplus 0 \oplus 1 \oplus 0 \oplus 0 = 0$; $P_2 = 1 \oplus 1 \oplus 1 \oplus 1 \oplus 0 = 0$
4. $P_4 = 0 \oplus 1 \oplus 1 \oplus 1 = 1$; $P_8 = 0 \oplus 1 \oplus 0 \oplus 1 = 0$
5. Code word **0011 0110 0101**
6. Received: 1s at positions 3, 4, 6, 7, 12
7. XOR them: $0011 \oplus 0100 \oplus 0110 \oplus 0111 \oplus 1100 = 1010$
8. Syndrome **10**: flip bit 10 → the original word ✓

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Break

10 minutes — we resume at the 1:00 mark

Next: ROM, PLA and PAL

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Review 4 — ROM sizing

§7.5

REVIEW

Give the ROM size for (a) a BCD-to-excess-3 converter, (b) a 4-bit adder with carry in, (c) any 3 functions of 6 variables.

Circuit	Inputs	Outputs	ROM
BCD to excess-3	4	4	16 × 4
4-bit adder with carry in	9	5	512 × 5
3 functions of 6 variables	6	3	64 × 3

1. Rule: k inputs, n outputs → a $2^k \times n$ ROM
2. BCD to excess-3: 4 in, 4 out → $2^4 \times 4 = $ **16 × 4**
3. Adder: $4 + 4 + 1 = 9$ in; 4 sum + carry = 5 out → **512 × 5**
4. 3 functions of 6 variables: $2^6 \times 3 = $ **64 × 3**

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Review 5 — PLA with shared terms

§7.6

REVIEW

Implement $F_1 = \Sigma m(1,2,3,7)$ and $F_2 = \Sigma m(0,1,5,7)$ in a PLA with the fewest product terms.

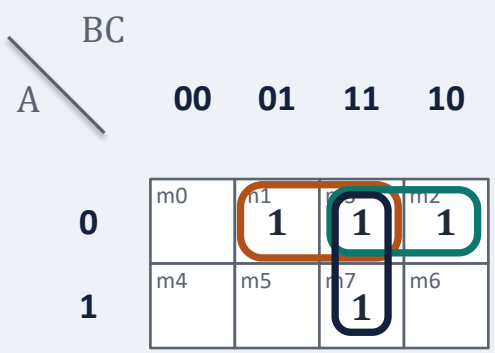
Choice	Products	Count
F1, F2	A'B, A'B', A'C, AC, BC	5
F1, F2'	A'B, A'C, AC', BC	4
F1', F2	A'B', AB', AC, AC', B'C'	5
F1', F2'	A'B, AB', AC', B'C'	4

1. Minimize each output and its complement
2. $F_1 = A'C + A'B + BC;\ F_1' = B'C' + AB' + AC'$
3. $F_2 = A'B' + AC;\ F_2' = A'B + AC'$
4. Count distinct products for each pair (table)
5. F_1 with F_2' shares $A'B$: **4 products**
6. Program output 2 as C (complement)
7. True forms of both would need 5

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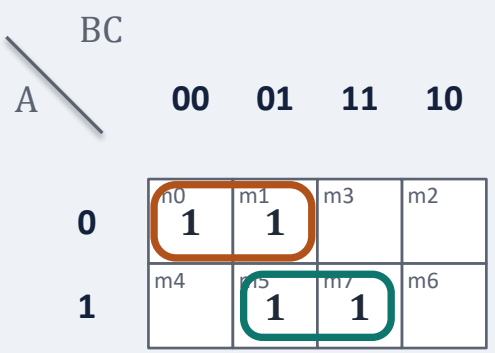
Review 5 — The four K-maps

§7.6

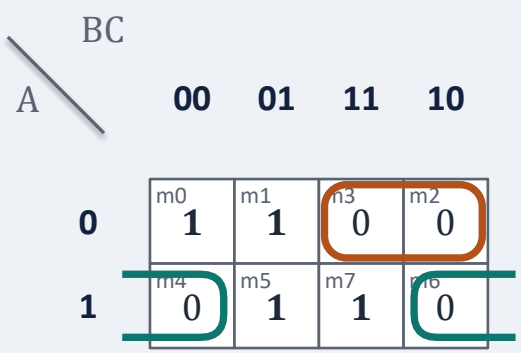


$F_1 = A'C + A'B + BC$

$F_1' = B'C' + AB' + AC'$



$F_2 = A'B' + AC$



$F_2' = A'B + AC'$

1. Group the 1s: three pairs
2. $A'B, A'C, BC$
3. Group the 0s: three pairs
4. Shares nothing with F_2
5. Group the 1s: two pairs
6. Shares nothing with F_1
7. Group the 0s: two pairs
8. $A'B$ is also in F_1

Best pair: F_1 and F_2' share $A'B$ → 4 products; invert output 2.

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Common errors in this unit

§7.5

Memory

Confusing words with bits: 2K × 8 is 2048 words, 16,384 bits.

Rounding clock periods down instead of up.

Sending the high address bits to every chip instead of to the chip-select decoder.

Hamming

Putting data bits in positions 1, 2, 4, 8.

Using odd parity when the code is even.

Programmable logic

Minimizing literals for a PLA: minimize distinct product terms instead.

Forgetting to try F' for each output.

Sharing terms in a PAL: its OR array is fixed.

Sizing a ROM from the minimized function: a ROM stores every row.

When in doubt, write the truth table first.

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Practice set — 15 minutes

§7.2

YOUR TURN

≈ 15 min

1. How many bits does a 4K × 4 memory store? How many address lines?
2. Encode the data 0110 as a 7-bit Hamming code word.
3. The 7-bit word 1100111 is received. Syndrome? Corrected word?
4. Which device has a fixed AND array and a programmable OR array?
5. Why does an FPGA lose its configuration at power-off?

Work alone, as on the exam.

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Practice set — Solutions

§7.2

SOLUTION

1. $4096 \times 4 =$ **16,384 bits**; $4096 = 2^{12}$ → **12** lines

2. Data 0110 → bit 3 = 0, 5 = 1, 6 = 1, 7 = 0

3. $P_1 = 0 \oplus 1 \oplus 0 = 1;\ P_2 = 0 \oplus 1 \oplus 0 = 1$

4. $P_4 = 1 \oplus 1 \oplus 0 = 0$ → **1100110**

5. 3. 1100111: 1s at positions 1, 2, 5, 6, 7

6. XOR: $001 \oplus 010 \oplus 101 \oplus 110 \oplus 111 = 111 = 7$

7. Flip bit 7 → **1100110**

8. 4. Fixed AND array + programmable OR: the **ROM**

9. 5. Its configuration is in SRAM cells (volatile)

10. It reloads from flash at power-up

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The comprehensive final

Unit	Topics
Numbers, codes	bases, complements, signed arithmetic, BCD, Gray, parity
Boolean algebra	theorems, canonical forms, K-maps, tabular method, NAND/NOR
Combinational	analysis, design, adders, BCD, comparators, decoders, MUXes
Sequential	latches, flip-flops, analysis, reduction, design
Registers, counters	shift registers, ripple and synchronous counters
Memory, PLDs	RAM, decoding, Hamming, ROM, PLA, PAL

Lecture 29 reviews the whole course with one worked problem per unit.

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Study plan

Two weeks out

Redo every Your turn and practice set, one unit per day.

Rework the problems you missed on Exams 1–3.

Last three days

Timed practice: one mixed problem set per day.

Write the summary equations from memory.

On the exam

Name the procedure (ANALYSIS, DESIGN) and number the steps.

State the MSB and every assignment you choose.

Check each answer on one row or by converting back.

Bring

Pencil, eraser, calculator.

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Key takeaways

Capacity

$2^k \times n$ bits

Expansion

high bits → chip selects

Wait states

$\lceil t_{cycle}/T \rceil$

Hamming

$2^k \geq n + k + 1$

ROM / PLA / PAL

OR / both / AND programmable

PLA trick

share terms via F'

Memory and PLDs are decoders and arrays you already know.

This afternoon's lab: Lab L — 2 × 2-bit RAM

Next: comprehensive final review

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